

Tinkering & Mentoring - VACC 201



Problem Identification, Ideation & Initial Design



Production of Bioethanol

Date (30/01/2026)

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Faculty Mentor Name: Dr. Arti Hansda

Team No.: 10

Semester: 2nd

Program: B.sc Biotechnology

School: School of Science

Student Team

Name	Enrollment No.	Key Responsibility
MANN PATEL	25SC02056	Taking care of requirements and availability responsibilities
NANDINI PATEL	25SC02057	Lab Responsibilities
PAL PATEL	25SC02058	Lab Responsibilities
RUTVI PATEL	25SC02059	Material and Status management of Project
SHRADDHA PATEL	25SC02060	Material Management
VRAJ PATEL	25SC02061	Taking care of requirements and availability responsibilities

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Faculty Mentor

Name	Designation	Department, School
Dr. ARTI HANSDA	Assistant Professor	Life Science, School of Science

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Brief of Tinkering Project

- **Title of Tinkering Project: Production Of Bioethanol**
- **Sector: Biotechnology**
- **Brief of Tinkering Project:** The production of bioethanol is an important biotechnological process that converts biomass into a renewable fuel through fermentation. Due to the depletion of fossil fuels and increasing environmental pollution , bioethanol serves as a sustainable alternative energy source. This project explores bioethanol and its relevance in reducing carbon emissions.

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Objective

- To produce bioethanol from renewable biological source such as sugarcane juice, corn or agricultural waste through fermentation.
- To study the role of microorganisms like *Saccharomyces cerevisiae* in converting sugars into ethanol.
- To optimize fermentation conditions such as temprature , pH, and sugar concentration for maximum ethanol yield.
- To develop an eco-friendly and sustainable alternative fuel that can reduce dependency on fossil fuels.

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USP/ Novelty/ Unique Points of Tinkering Project

- Bioethanol is a renewable and eco-friendly alternative to petrol.
- It significantly reduces greenhouse gas and carbon monoxide emissions.
- Produced from organic and effective agricultural waste.
- Supports sustainable energy by converting biomass into clean fuel.
- Blending ethanol with gasoline improves engine efficiency and fuel economy.

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Project Timeline & Milestones

Timeline : 1 – 2 Months

PHASE	KEY PROCESS	MILESTONE ACHIEVED
Phase 1	Literature survey and project planning	Process workflow finalized
Phase 2	Collection and pre-treatment of biomass	Raw substrate prepared
Phase 3	Fermentation using Yeast (<i>Saccharomyces cerevisiae</i>)	Ethanol produced in broth
Phase 4	Distillation and purification of ethanol	Concentrated bioethanol obtained
Phase 5	Testing (alcohol percentage and efficiency)	Final Project report completed

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➤➤ **Thank You..**